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```
> bootcamp.learn()
```

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JomCode Coding Bootcamp

CHAPTER 03

Values, types, operators and strings

```
learn / type / run / explain
```

Plain meaning and technical explanation

PLAIN MEANING

A post contains text, counts, yes/no choices and sometimes a missing image.

TECHNICAL EXPLANATION

Variables refer to values. int, float, str, bool and None represent different kinds of data. Operators and methods depend on type.

REAL-WORLD USE

Model post fields and format previews before validating a form.

QUICK TIP

Change one value, predict the result, then run again.

INSTALL / RUN

No packages. Windows: py workbook.py. macOS: python3 workbook.py.

Worked example 1: Post field types

workbook.py

```
01    title = "Hello"
02    views = 12
03    published = True
04    image_path = None
05    print(type(title).__name__)
06    print(type(views).__name__)
07    print(type(published).__name__)
08    print(type(image_path).__name__)
```

PRO TIP Use meaningful variable names.

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Run and explain example 1

EXPECTED RESULT

str
int
bool
NoneType

WHY IT BEHAVES THIS WAY

None represents absence. An empty string still has type str.

Worked example 2: Size and precedence

workbook.py

```
size_bytes = 1_500_000
size_mb = size_bytes / 1_000_000
print(size_mb)
print(2 + 3 * 4)
print((2 + 3) * 4)
print(23 // 5, 23 % 5)
```

PRO TIP Decimal MB uses 1,000,000 bytes; MiB uses 1,048,576.

Run and explain example 2

EXPECTED RESULT

1.5
14
20
4 3

WHY IT BEHAVES THIS WAY

Multiplication happens before addition. // counts full groups and % gives the remainder.

Worked example 3: Clean and preview text

workbook.py

```
01    title = "  My  Story  "
02    clean_title = title.strip()
03    print(clean_title)
04    print(clean_title[:5])
05    print("Go"[:5])
06    print(f"{clean_title}: {len(clean_title)} characters")
```

PRO TIP Internal spaces remain unchanged.

Run and explain example 3

EXPECTED RESULT

My Story
My S
Go
My Story: 9 characters

WHY IT BEHAVES THIS WAY

strip() removes edge whitespace. Slicing stops safely at the end of short strings.

Worked example 4: Known-valid input conversion

workbook.py

```
01    first = "12"
02    second = "8"
03    print(first + second)
04    total = int(first) + int(second)
05    print(total)
06    print(f"{total} posts published")
```

PRO TIP Conversion can fail. Chapter 9 handles invalid input.

Run and explain example 4

EXPECTED RESULT

128
20
20 posts published

WHY IT BEHAVES THIS WAY

Adding strings concatenates text. Convert valid numeric text before arithmetic.

Exercise 1: Post summary

YOUR TASK

Create title, views, published and missing-image fields. Print a trimmed five-character preview and total views.

CHECK THESE CASES

"Go" stays "Go". "12" and "8" total 20. Missing image is None.

HINT AFTER A TRY

Use strip(), slicing, int() and an f-string.

Exercise 2: File-size report

YOUR TASK

Normalize an email and report 2,500,000 bytes in decimal MB. Preserve internal title spacing.

CHECK THESE CASES

Email becomes ada@example.com. Size is 2.5 MB. " My Story " keeps its two internal spaces.

HINT AFTER A TRY

Use strip(), lower() and division.

Checkpoint before moving on

01 Run and explain all four worked examples. Match their actual results to the expected behavior.

02 Attempt both exercises before reading the solutions. Verify every acceptance case.

03 Explain how this chapter supports Postlight: Model post fields and format previews before validating a form.

Official reference